

HR Attrition Analysis

This analysis was conducted to identify the key factors that influence employee attrition and to help the HR team make informed decisions to improve employee retention. The dataset contains information about employees, including their job roles, salaries, work-life balance, business travel, and other work-related factors.

The overall attrition rate in the company is **16.12%**, indicating that approximately one out of every six employees leaves the organization. The analysis shows that the **Sales Department** experiences the highest attrition rate (20.63%), making it the department that requires the most immediate attention. Among all job roles, **Sales Representatives** have the highest attrition rate at **39.76%**, followed by **Laboratory Technicians** and **Human Resources** employees.

Several factors were found to have a strong influence on employee attrition. Employees who work **overtime**, travel **frequently for business**, or work as **Laboratory Technicians** are more likely to leave the company. Employees with lower monthly incomes also tend to leave more often, but salary is not the only reason for attrition. Work-life balance, job role, and travel requirements play an equally important role in employee retention.

Three machine learning models were developed to predict employee attrition: Logistic Regression, Random Forest, and Gradient Boosting. Among these, **Logistic Regression** performed best for this problem because it achieved the highest ROC-AUC score and was more effective at identifying employees who were likely to leave. Its simplicity also makes it easier for HR teams to understand and use in decision-making.

Based on these findings, it is recommended that the company reduce excessive overtime, improve work-life balance, and provide career development opportunities for employees in high-risk roles, particularly Sales Representatives and Laboratory Technicians. Regular retention discussions and employee engagement programs should also be introduced for employees in departments with high turnover.

While the model provides useful insights, it should be used as a decision-support tool rather than the only basis for HR decisions. Employee attrition can also be influenced by personal circumstances, workplace culture, and external opportunities, which are not captured in the available data.